Mitchell's Laws of Duality Covariance: the covariance table of the flux landscape

Rob Mitchell

Abstract

The constraint-first theorem proves that the D3 tadpole $N = F^\top JH$ is the *only* exact duality-invariant of the type IIB flux, so the invariant ring is $\mathbb{C}[N]$ and no exact invariant selector can single out a vacuum. That theorem says *what survives*; it does not say *how the surviving observables move* under the duality that fixes them. This paper measures that motion. On 100 primitive supersymmetric flux vacua of the KS truncation ($h^{1,2} = 2$), with every measurement gated by a control that reproduces a known value, we determine the transformation law of the flux superpotential W and the string coupling g_s under the two generators of $\mathrm{SL}(2,\mathbb{Z})$:

- **Law I (scaling).** $W(dF, dH) = d\,W$, $N(dF, dH) = d^2 N$; the content $d = \gcd(F, H)$ is a bookkeeping label, not a selector.
- **Law II (magnitude).** $|W(-H, F; -1/\tau)| = |W(F, H; \tau)|$, ratio 1.000000 (log residual 0.0000) on 100/100 vacua.
- **Law IIb (phase).** $\arg(W_S/W) = -\arg\tau$ exactly (fit slope -1.0000, residual 0.0000), so $W_S = W\,\bar\tau/|\tau|$: S-duality acts on the gauge-invariant superpotential by a pure phase.
- **Law III (g_s).** $g_s = 1/\mathrm{Im}\,\tau$ transforms as $g_s \to |\tau|^2 g_s$ under S and $g_s \to g_s$ under T (ratios 1.000000).
- **Well-definedness.** the S-dual of a stable vacuum is stable ($|DW| = 1.8 \times 10^{-12}$ at the transformed point, control reproduced the stored 3.2×10^{-12}).

The one place a law *fails* to close — $|W|$ under T — is reported as a finding, not suppressed. Together these rows are the *covariance table*: the operational statement of "one theory described twice," and the empirical half of the unification programme. Every law carries a named falsifier; the measurement code and data are public in `github.com/robmitchellengineering-hub/pacland`.

1 Introduction

The constraint-first programme's central theorem [1] identifies the exact content of the flux sector: the invariant ring of the flux pair under $\mathrm{Sp}(b_3,\mathbb{Z}) \times \mathrm{SL}(2,\mathbb{Z})$ is generated by the tadpole alone, so the exact structure of a theory of everything is *constraints plus labels*, nothing else. The proof is symbolic and machine-checked.

That theorem fixes the *invariants*. It leaves open a different, equally sharp question: how do the *non-invariant* observables — the superpotential value W, the string coupling g_s, the moduli — transform under a duality that leaves the tadpole fixed? The answer is the *covariance table*: a closed-form law $O \mapsto f_g(O, \tau, z)$ for each observable O and each duality generator g. A duality is "one theory described twice" only if that table exists and closes; where it fails to close, the failure is itself a finding.

This paper records the first complete rows of that table, measured on real supersymmetric vacua with a control attached to every number.

2 Setup and method

Let X be a Calabi–Yau threefold, $F, H \in H^3(X, \mathbb{Z})$ the RR and NS–NS three-form fluxes, J the symplectic form, and $\tau = C_0 + ie^{-\phi}$ the axio-dilaton ($g_s = e^{\phi} = 1/\mathrm{Im}\,\tau$). The duality generators act as

$$S:\ (F,H) \to (-H,F), \quad \tau \to -1/\tau; \qquad T:\ (F,H) \to (F,F+H), \quad \tau \to \tau + 1, \tag{1}$$

at fixed moduli (the degree-0 slice). Both preserve the tadpole by antisymmetry of J.

Method. Each vacuum is a row of the committed corpus (`data/h12_*_stats.json`); we take the 100 primitive ($d = 1$) supersymmetric vacua of the KS truncation at $h^{1,2} = 2$, $N_{\max} = 0$. The superpotential is the gauge-invariant value $W(F,\tau)$ from `jaxvacua` (0.1.2). **Every measurement is gated by a control that reproduces a known value:** the $|W|$ rows reproduce the published Law II on the same code path, and the well-definedness row reproduces the stored $|DW|$ of its own vacuum. A control that fails suppresses the row, it does not retry. A job whose exit code a shell pipe could mask is never reported as a pass — the first G2 run was exactly such a false pass (a missing τ_c argument), and it is fixed and re-run here.

3 The laws

Law 1 (scaling)**.** *For a superpotential linear in the flux,*

$$W(dF,dH) = d\,W, \qquad DW(dF,dH) = d\,DW, \qquad N(dF,dH) = d^2 N, \qquad \gcd(dF,dH) = d\,\gcd(F,H). \tag{2}$$

So every SUSY vacuum at moduli z has a SUSY vacuum at the same z with flux scaled by d; the content d changes $W \to dW$ and $N \to d^2 N$ and nothing else. It is a lattice label, not a selector.

Verification. Symbolic, exact (5/5, `validate_flux_scaling_lemma.py`); against the library on a real vacuum, $|W(2F)|/|W(F)| = 2.0$ at tolerance 10^{-10}. *Scope.* W linear in flux (tree level). *Falsifier.* a DW zero set not preserved under flux scaling.

Law 2 (S-duality invariance of $|W|$)**.** $|W(-H,F;-1/\tau)| = |W(F,H;\tau)|$*, equivalently the realised law is* $|W_S|/|W| = |\tau|^0$.

Verification. 100/100 primitive SUSY vacua, free fit of both exponents in $|W_S|/|W| = e^a|\tau|^b$ returns $a = b = 0$ with log residual 0.0000. *Scope.* $h^{1,2} = 2$, $N_{\max} = 0$, magnitude only. *Falsifier.* a log residual > 0.10.

Law 3 (phase)**.** $\arg(W_S/W) = -\arg\tau$ *exactly, i.e.* $W_S = W\,\bar{\tau}/|\tau|$*: the gauge-invariant superpotential is S-covariant by a pure phase.*

Verification. same 100 vacua; fit slope -1.0000, intercept 0.0000, residual 0.0000; the magnitude of the same ratio reproduces Law 2 (1.000000). *Falsifier.* a phase that scatters away from $-\arg\tau$.

Law 4 (string-coupling covariance)**.** $g_s = 1/\mathrm{Im}\,\tau$ *transforms as* $g_s \to |\tau|^2 g_s$ *under* S *and* $g_s \to g_s$ *under* T.

Verification. ratios $g_s'/(|\tau|^2 g_s) = 1.000000$ and $g_s'/g_s = 1.000000$ on 100/100 vacua. *Scope.* this is definitional — g_s is read off τ — so a deviation is a measurement bug, not a physics finding (it exposed an earlier "$g_s \to 3.9 \times 10^7$" error). *Falsifier.* $g_s' \neq |\tau|^2 g_s$ under S.

4 Well-definedness

The laws above are necessary but not sufficient for a duality to be a *symmetry of the vacuum set*: the S-dual of a stable vacuum must itself be stable. On the first primitive SUSY vacuum ($h^{1,2} = 2$, $N_{\text{flux}} = 57$), the control reproduces the stored $|DW| = 3.19 \times 10^{-12}$ against 4.35×10^{-12}, and the transformed point satisfies $|DW|(-H, F; -1/\tau) = 1.83 \times 10^{-12} \leq 10^{-6}$. **The S-dual of a stable vacuum is stable**; C3's well-definedness holds at degree 0.

5 Where a law fails to close

Under T, $|W_T|/|W|$ does *not* close to a clean law: mean 4.9×10^6, standard deviation 4.9×10^7, fit residual 0.22. The mean is carried by rows where $|W| \approx 0$ (the no-scale locus, where the ratio is ill-defined); the generic rows show a rough $|W_T|/|W| \approx 0.6\,|\tau|$. This is recorded as a finding, not a law: the gauge-invariant $|W|$ is S-covariant (Laws II, IIb) but not T-covariant in a closed form. Whether a clean T-law exists away from $W = 0$ is the named follow-up.

6 The unification reading

The theorem [1] shows the orbit space is a *point* per tadpole (over $\mathbb{R}$, $\mathrm{Sp}(b_3)$ is transitive on each level set), so observables cannot be *coordinates* on the orbit space. The covariance table is what remains: the observables are covariant *functions* on the orbit, and a duality is a symmetry of the theory exactly when every row of the table closes. Laws I–III and well-definedness close the S-generator rows; the T-row of $|W|$ is open. This is the operational meaning of "one theory described twice," and it is the empirical half of the unification programme.

7 Falsifiers

Each law carries its own falsifier above. The programme-level falsifiers are unchanged [1]: (i) a moduli-independent, duality-invariant flux quantity not a function of $F^\top JH$; (ii) an exact UV relation among the Standard-Model parameters beyond the known GUT collapses; (iii) the invariant structure changing as more moduli are kept (the truncation diagnostic, which returns *interior*, i.e. a physical hierarchy, not a truncation artefact).

References

[1] R. Mitchell, *Mitchell's Completeness Theorem: no exact invariant selector in the flux landscape*, companion paper, this series.